

TAUPO, New Zealand

WMO: 932450

Lat: 38.742S

Lon: 176.081E

Elev: 1317

StdP: 14.01

Time Zone: 12.00 (NZL)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.8	29.8	25.3	20.3	29.6	27.2	22.2	31.7	27.5	48.1	25.0	47.6	3.5	80	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.9	76.8	63.6	74.5	62.8	72.4	62.0	67.0	72.7	65.5	71.0	64.0	69.5	7.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	97.0	69.7	63.3	91.5	68.3	61.8	86.6	67.1	32.2	72.8	31.0	71.1	29.9	69.5	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.6	21.7	19.3	DB	24.5	81.5	0.9	2.4	23.9	83.2	23.3	84.6	22.8	86.0	22.2	87.7
			WB	23.9	69.2	1.1	1.5	23.1	70.2	22.5	71.1	21.9	71.9	21.1	72.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	62.9	63.1	59.6	53.9	49.8	45.7	44.2	45.4	48.4	51.8	55.2	59.9
	DBStd	8.05	4.52	4.46	4.45	4.80	5.31	5.42	4.83	4.28	4.19	4.35	4.67	4.39
	HDD50	701	0	0	1	16	71	150	187	151	79	34	10	2
	HDD65	4343	92	80	170	332	470	580	645	609	499	409	294	163
	CDD50	1898	401	367	300	135	66	20	7	7	30	89	167	309
	CDD65	67	29	27	4	0	0	0	0	0	0	0	1	6
	CDH74	247	131	81	9	0	0	0	0	0	0	0	8	18
	CDH80	14	9	4	0	0	0	0	0	0	0	0	1	0
(14) Wind	WSAvg	8.1	8.4	7.7	7.5	7.1	7.4	7.8	7.6	7.5	8.4	9.5	9.4	8.5
(15) Precipitation	PrecAvg	49.9	3.6	3.4	3.5	3.9	4.2	4.7	5.0	4.9	4.3	4.1	3.7	4.6
	PrecMax	67.6	11.4	9.3	7.5	7.8	7.0	8.5	11.6	9.2	7.6	8.1	7.0	8.1
	PrecMin	38.8	1.1	0.7	1.4	1.2	1.3	2.0	1.5	2.2	1.6	1.3	1.7	0.6
	PrecStd	7.0	2.2	2.0	1.6	1.8	1.5	1.5	2.3	1.8	1.6	1.5	1.5	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.8	79.7	75.2	69.1	64.2	59.6	57.1	57.4	62.1	66.6	74.7	76.7
		MCWB	65.4	65.6	62.2	61.4	59.4	56.7	53.8	52.5	54.7	55.5	60.3	62.9
	2%	DB	77.4	76.4	72.3	66.0	61.8	57.0	54.7	55.5	59.3	63.4	70.1	73.0
		MCWB	63.7	64.2	61.4	59.3	58.1	54.4	51.4	50.8	53.3	55.2	58.9	61.6
	5%	DB	74.8	74.2	70.0	64.1	59.9	55.3	53.1	53.8	57.6	61.5	66.9	70.7
		MCWB	63.1	63.3	60.7	58.4	56.5	52.6	49.9	49.6	52.4	54.4	57.7	60.7
	10%	DB	72.4	72.0	68.0	62.3	58.2	53.8	51.7	52.3	56.0	59.6	64.0	68.4
		MCWB	62.2	62.5	60.2	57.2	54.8	51.0	48.4	48.6	51.5	53.7	56.3	59.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.1	69.2	66.2	63.8	61.1	57.5	55.1	54.4	56.7	58.5	63.5	66.1
		MCDB	76.2	74.4	70.4	66.6	62.8	58.8	56.7	56.2	59.3	63.3	69.8	71.7
	2%	WB	67.1	67.2	64.2	61.3	59.0	54.9	52.2	52.0	54.8	56.8	60.9	64.0
		MCDB	73.3	72.7	68.5	64.3	61.1	56.4	54.0	54.2	57.8	60.7	66.9	69.3
	5%	WB	65.6	65.6	62.7	59.6	57.1	53.0	50.4	50.4	53.4	55.6	59.0	62.5
		MCDB	71.5	71.1	67.4	63.0	59.6	54.8	52.6	52.9	56.8	59.9	65.0	67.9
	10%	WB	63.8	64.1	61.3	58.0	55.1	51.3	48.9	49.2	51.9	54.4	57.4	61.1
		MCDB	69.7	69.8	66.5	61.6	57.7	53.4	51.2	51.7	55.2	58.8	62.9	66.6
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	16.9	16.6	16.0	14.1	13.1	13.3	13.7	14.5	14.8	16.1	15.3
		MCDBR	21.4	19.9	18.7	17.0	13.2	10.5	11.3	14.1	16.0	17.6	20.7	19.1
		MCWBR	10.3	9.6	9.8	11.0	9.6	8.6	8.5	10.1	10.8	11.1	11.5	10.1
	5% WB	MCDBR	16.1	14.7	14.1	12.4	10.7	9.0	9.3	10.8	12.7	14.1	17.6	14.5
		MCWBR	9.2	8.3	8.7	9.0	8.5	8.2	7.7	8.5	9.5	9.9	11.0	8.9
(40) Clear-Sky Solar Irradiance	taub		0.326	0.327	0.306	0.294	0.281	0.276	0.274	0.283	0.307	0.317	0.316	0.329
	taud		2.571	2.575	2.640	2.664	2.677	2.663	2.671	2.645	2.547	2.531	2.556	2.558
	Ebn at Noon		319	311	306	291	276	267	276	290	300	311	320	319
	Edh at Noon		33	32	27	23	20	18	19	23	29	33	34	34
(44) All-Sky Solar Radiation	RadAvg		2042	1763	1434	1000	682	540	613	835	1146	1515	1834	1893
	RadStd		150	179	89	72	46	47	53	61	98	109	147	169

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	+1.48	+1.80	N/A	N/A	N/A	N/A	-90	-211	N/A	N/A
(47) Regional (6 neighbors)	+0.49	+0.65	+1.03	N/A	N/A	N/A	N/A	-43	-164	+149	N/A

Nomenclature: See separate page